Model Performance Comparison

Airbnb Price Prediction - Regression

ITAI 1371 - Classic Machine Learning

Student: Mary Balemba

Final Project - August 2026

Dataset: Airbnb Open Data

1. Evaluation Overview

The final project evaluates five required regression algorithms, an average ensemble of the three strongest validation models, and Bayesian Ridge. The cleaned Airbnb dataset was split into 70% training, 15% validation, and 15% test data. Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 were used to compare performance. Lower MAE and MSE are preferred, while higher R^2 indicates stronger explanatory performance.

2. Validation and Test Comparison

Model	Val MAE	Val MSE	Val R^2	Test MAE	Test MSE	Test R^2
Random Forest Regressor	**224.8594**	**76,542.1309**	**0.3031**	**223.6204**	**76,230.0658**	**0.3022**
Average Ensemble (Top 3)	265.0385	94,914.7498	0.1358	264.3793	94,434.8006	0.1356
Gradient Boosting Regressor	285.9776	109,639.0844	0.0017	285.4698	108,927.2936	0.0029
Bayesian Ridge	286.3809	109,854.4222	-0.0003	285.9414	109,253.1262	-0.0001
Linear Regression	286.4297	109,880.4135	-0.0005	285.9616	109,286.6749	-0.0004
K-Nearest Neighbors Regressor	301.4466	128,800.5200	-0.1728	302.7520	129,703.3286	-0.1873
Decision Tree Regressor	252.5198	144,582.2790	-0.3165	251.5091	144,107.3922	-0.3191

Random Forest Regressor produced the strongest overall test performance with MAE 223.6204, MSE 76,230.0658, and R^2 0.3022. Its validation R^2 of 0.3031 was nearly identical to its test R^2, which indicates stable generalization.

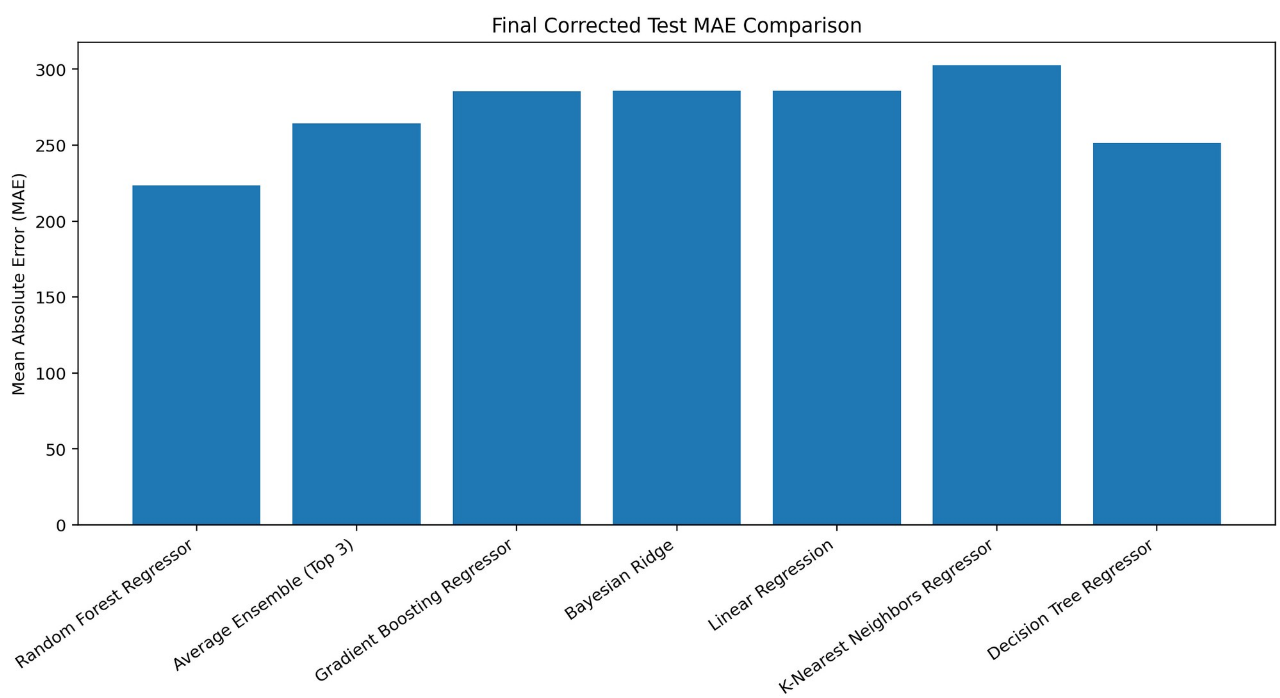

Figure 1. Test MAE comparison. Lower values are better.

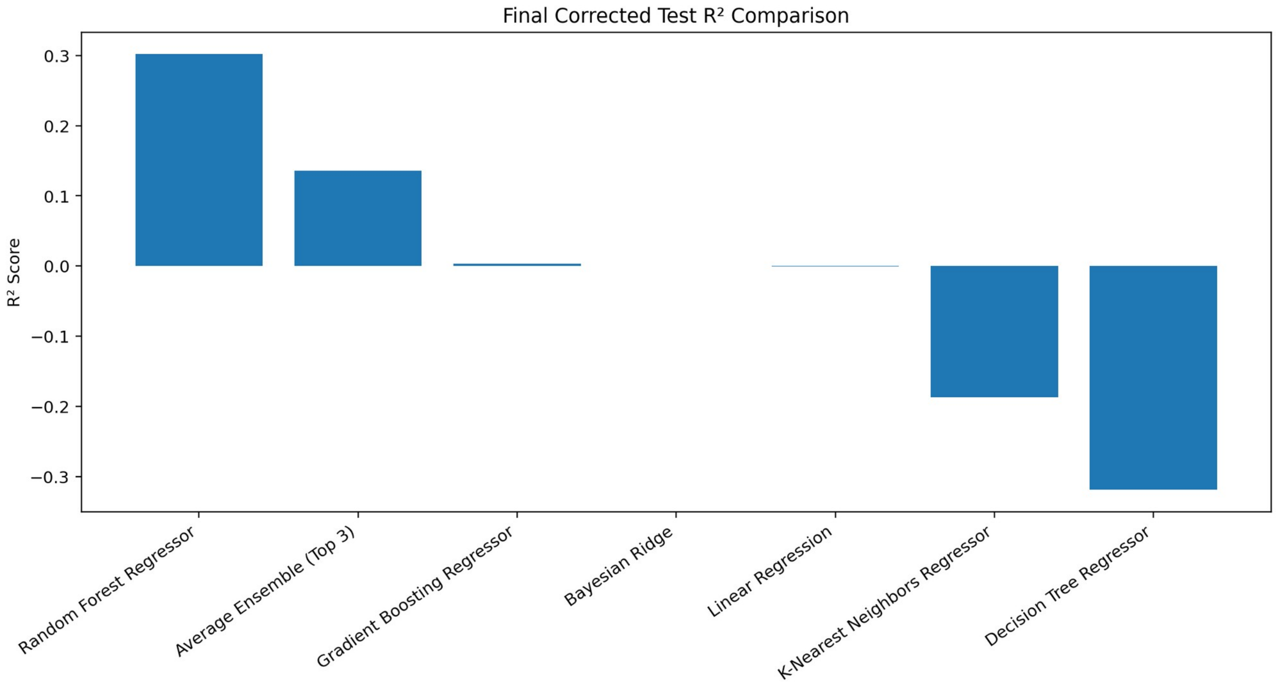

Figure 2. Test R^2 comparison. Random Forest produced the highest value.

3. Model Discussion

Linear Regression

Linear Regression produced validation MAE 286.4297, MSE 109,880.4135, and R^2 -0.0005. Test R^2 was -0.0004. The near-zero R^2 shows that the remaining predictors did not explain Airbnb price well through a simple linear relationship.

Decision Tree Regressor

Decision Tree produced validation R^2 -0.3165 and test R^2 -0.3191. Its test MSE of 144,107.3922 was the highest of all evaluated approaches, indicating some large prediction errors and weak generalization.

Random Forest Regressor

Random Forest was the best model. It achieved validation MAE 224.8594, MSE 76,542.1309, and R^2 0.3031; test MAE 223.6204, MSE 76,230.0658, and R^2 0.3022. The model likely benefited from combining many trees and capturing nonlinear interactions among location, review, availability, and listing characteristics.

Gradient Boosting Regressor

Gradient Boosting achieved validation R^2 0.0017 and test R^2 0.0029. It performed slightly above the mean baseline but explained very little price variation with the selected configuration.

K-Nearest Neighbors Regressor

KNN achieved test MAE 302.7520 and R^2 -0.1873. Its negative R^2 and highest test MAE suggest that listings close in standardized feature space were not necessarily close in price.

Average Ensemble (Top 3)

The three best validation models were Random Forest, Gradient Boosting, and Linear Regression. Averaging their predictions produced test R^2 0.1356. The ensemble outperformed several models but did not beat Random Forest because averaging in weaker predictions reduced the strongest model's advantage.

Bayesian Ridge

Bayesian Ridge achieved test MAE 285.9414 and R^2 -0.0001. It was weaker than the Average Ensemble and Random Forest, indicating that a regularized Bayesian linear model did not capture the remaining nonlinear price relationships effectively.

4. Best and Worst Models

Best model: Random Forest Regressor. It had the lowest test MAE and MSE and the highest test R^2. Worst overall model: Decision Tree Regressor based on its lowest test R^2 (-0.3191) and highest test MSE (144,107.3922). This also shows why multiple metrics should be considered: the Decision Tree did not have the highest MAE, but its squared errors and overall explanatory performance were poor.

5. Target Leakage Correction

The first modeling run produced an unusually high R^2 near 0.995. A sanity check found that `service_fee` correlated approximately 0.998 with `price`, while `fee_to_price_ratio` was derived from price. These variables were removed to prevent target leakage, and all models were retrained. The corrected scores are lower but provide a much more realistic and defensible estimate of performance.

6. Final Selection

Random Forest Regressor was selected as the final model. Its test R^2 of 0.3022 means it explained approximately 30.22% of the variation in Airbnb listing prices in the unseen test data. Its very similar validation and test results suggest consistent generalization.